ROLL NO: | | | | | | | | | | |

(To fill the roll no is compulsory)

EVEN SEMESTER EXAMINATION - 2024-25

B.TECH. (CSE)

Semester: IV

SUBJECT CODE: BCS402

DESIGN AND ANALYSIS OF ALGORITHMS

Time: 03:00 Hrs. **Maximum Marks: 50**

NOTE: THE QUESTION PAPER CONTAINS THREE SECTIONS. ATTEMPT ALL THE SECTIONS.

SECTION – A

NOTE: Attempt all parts of the following question. (10 Marks)

Q. 1. Define the following terms: (1 × 5 = 5 Marks)

i. Differentiate between Complete Binary Tree and Binary Tree.

ii. Differentiate between Greedy Technique and Dynamic programming.

iii. Solve the following recurrence using Master method:
$T(n) = 4T(n/3) + n^2$

iv. Name the sorting algorithm that is most practically used and also write its Time Complexity.

v. Write the name of graph traversing techniques and also write its time complexities.

Q. 2. Answer the following questions: (1 × 5 = 5 Marks)

i. Write the time complexity of quick sort in worst case.

ii. Discuss the term in-place in sort.

iii. Compare Time Complexity with Space Complexity.

iv. What are the characteristics of the algorithm?

v. Differentiate between Backtracking and Branch and Bound Techniques.

SECTION – B

NOTE:Attempt all questions. **(5×2×2 = 20 Marks)**

Q.3.(i)(a) Solve the recurrence

1. T (n) =3T (n/4) + cn^2 using recursion tree method.
2. T (n) = n + 2T (n/2) using Iteration method. (Given T(1)=1)

OR

(b) Given the four matrices P, Q, R, T of dimensions 5x4,4x6,6x2,2x7. Find the optimal sequence using chain matrix multiplication (CMM).

(ii)(a) What are single source shortest paths? Write down Dijkstra's algorithm for it.

OR

(b) Write Merge sort algorithm and sort the following sequence {23, 11, 5, 15, 68, 31, 4, 17} using merge sort.

Q.4.(i) (a) What do you understand by stable and unstable sorting? Sort the following sequence {25, 57, 48, 36, 12, 91, 86, 32} using heap sort.

OR

(b) Discuss knapsack problem with respect to dynamic programming approach. Find the optimal solution for given problem, *w* (weight set) = {5, 10, 15, 20} and *W* (Knapsack size) = 25 and *v* = {50, 60, 120, 100}.

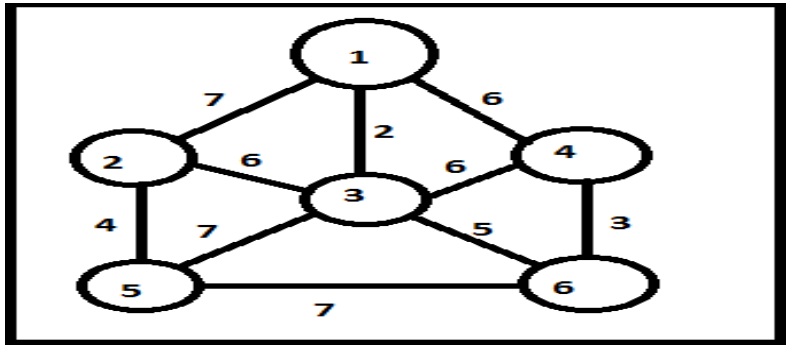
(ii)(a) Insert the following information

F,S,Q,K,C,L,H,T,V,W,M,R,N,P,A,B,X,Y,D,Z,E,G,I.

Into an empty B-tree with degree t=3.

OR

(b) What is Minimum Cost Spanning Tree? Explain prim's Algorithm and Find MST of the Graph. Also write its Time-Complexity.



SECTION – C

NOTE: Attempt all questions. **(10 × 2 = 20 Marks)**

Q.5.(a) What is Red-Black tree? Write an algorithm to insert a node in an empty red-black tree explain with suitable example.

OR

(b) Given the six items in the table below and a Knapsack with Weight 100, what is the solution to the Knapsack problem in all concepts. I.e. explain greedy all approaches and find the optimal solution.

ITEM ID	WEIGHT	VALUE	VALUE/WEIGHT
A	100	40	.4
B	50	35	.7
C	40	20	.5
D	20	4	.2
E	10	10	1
F	10	6	.6

Q.6.(a) Write down the Bellman Ford algorithm to solve the single source shortest path problem also write its time complexity.

OR

(b) Explain the method of finding Hamiltonian cycles in a graph using backtracking method with suitable example.